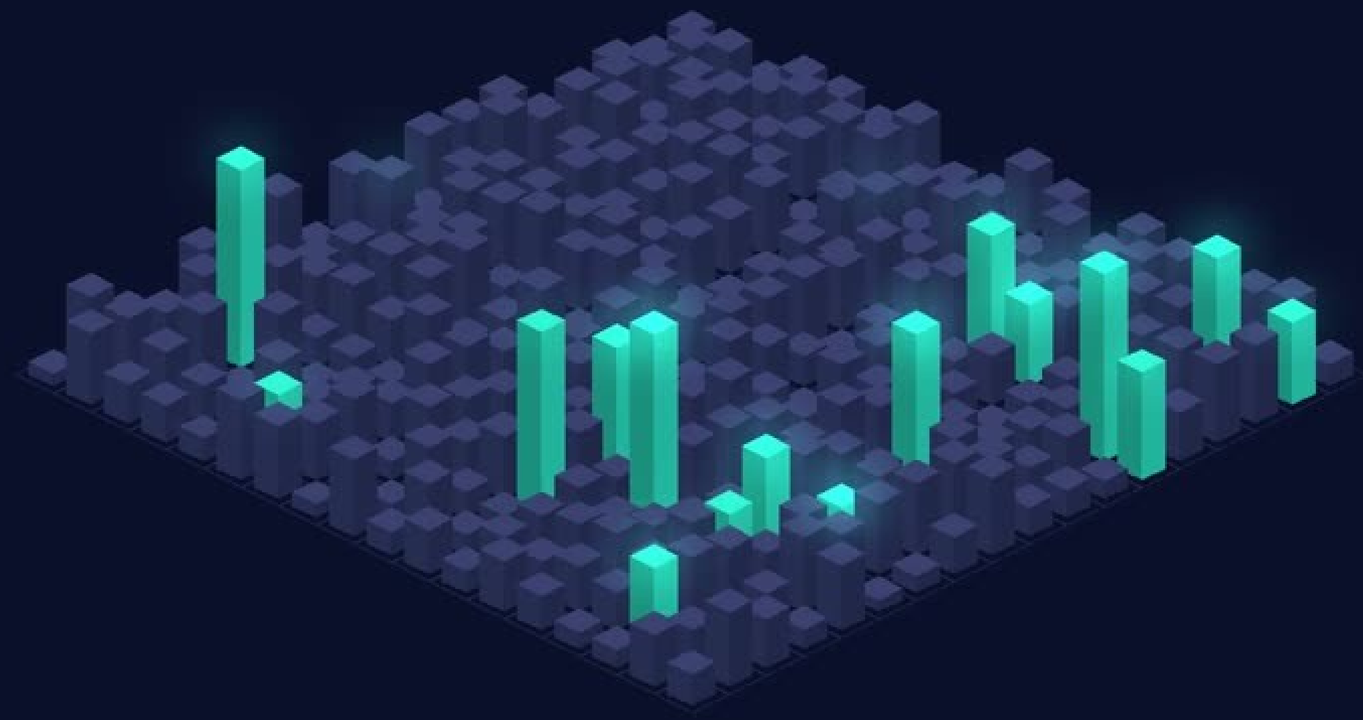
AGENTVERSE 2026 · BIOTECH TRACK · TEAM BACKPROP TO THE FUTURE

Which drug pairs should a lung cancer lab test next?

5,460

drug pairs

...but only a budget for 50



LungPair ranks drug combinations for non-small cell lung cancer labs: save time and money, change lives.

THE PROBLEM

Three bottlenecks between the lab and the patient

1

Too many pairs, too little budget

On osimertinib alone, 16 .7 month progression for EGFR+ NSCLC
With a second treatment, **extended** to **25.5** months.

The question is **which pair?**

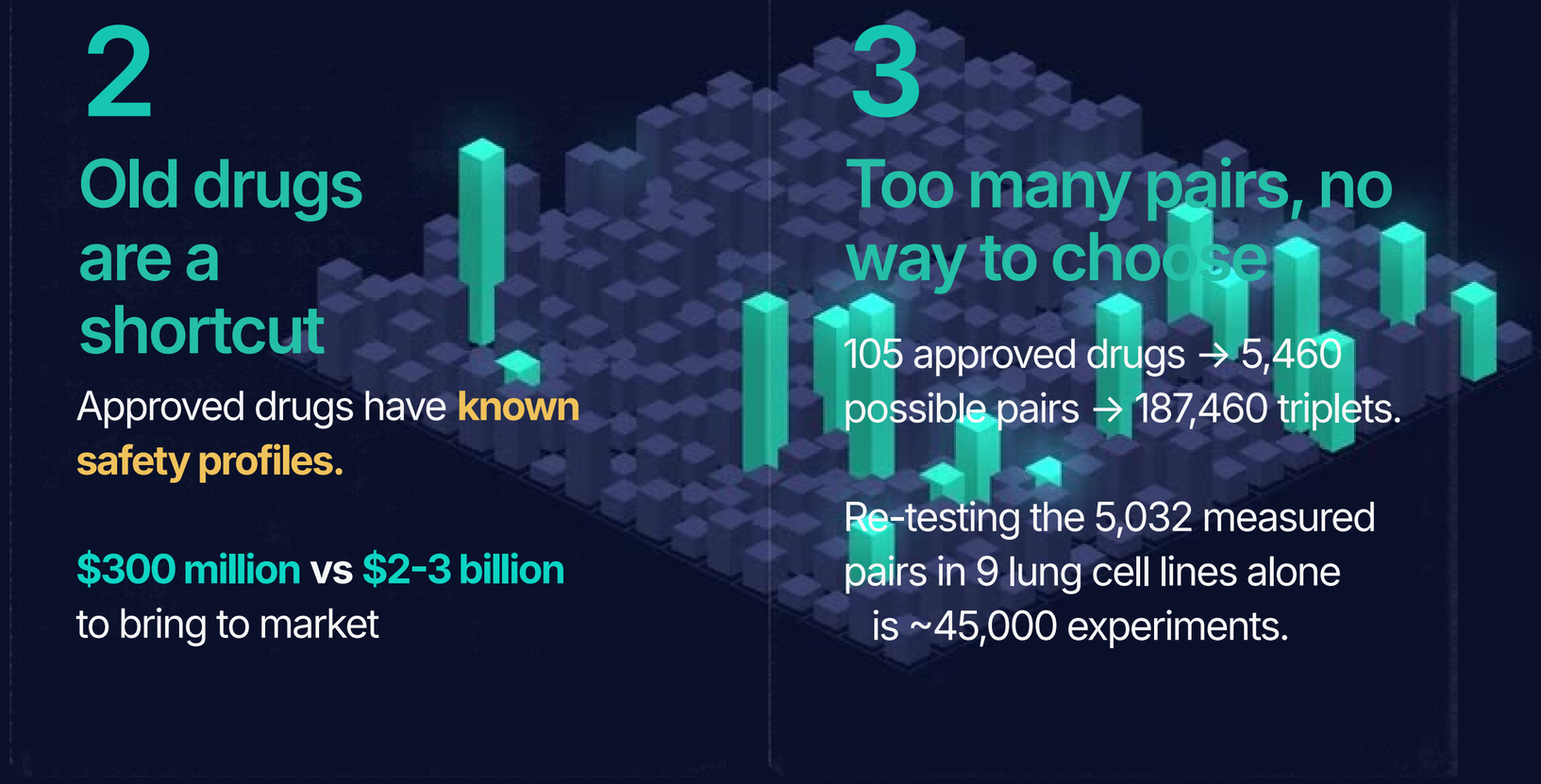
Labs currently choose
by literature review and intuition.

2

Old drugs are a shortcut

Approved drugs have **known safety profiles.**

$300 million vs **$2-3 billion**
to bring to market



3

Too many pairs, no way to choose

105 approved drugs → 5,460 possible pairs → 187,460 triplets.

Re-testing the 5,032 measured pairs in 9 lung cell lines alone is ~45,000 experiments.

Repurposing pathway: Kulkarni et al., Russ J Bioorg Chem 2023.

Cost: illustrative core-facility rate, 9 lung lines per pair.

WHO IT'S FOR

Built for lab researchers, not patients

WHO USES IT

Academic NSCLC labs

Lab technicians and translational teams

Screening cores and CROs (B2B)

WHAT IT GIVES THEM

A ranked **shortlist** of pairs to screen next

A **confidence tier** on every pair, with measured precision

A **refusal** when the evidence runs out

A **researcher** signs off every shortlist

Research use only. | A ranked pair is a hypothesis from cell-line data, not treatment advice.

THE PIPELINE

From public screens to a shortlist you can audit

Agents: Google Gemini. Models never compute a number; every score comes from the ranker.

1 Orchestrator
resolves drugs, calls the ranker

→

2 Ranker:
score, tier, 95% interval

→

3 Research agent:
mechanism + PubMed IDs

→

4 Checks:
5 code gates + checker

→

5 Researcher
signs off

→

6 Receipt:
every call logged

T4 refused, **returned as withheld.**

Held-out results (5 seeds)

0.89	0.61	0.82	0.71
precision@20	precision@50	AUPRC	MCC@50

precision@50 throughout

Hit prevalence 3.3%, so accuracy is not reported.

Confidence tiers (measured precision)

Tier	Definition	Measured	Lift vs baseline
T	pair measured in another cancer	0.89	24.0x
T1	both drugs screened, pair untested	0.34	11.0x
T2	new drugs, monotherapy known	0.06	3.0x
T3/T4	no evidence	0.04	refused

Sources: NCI DTP NCI-ALMANAC (ComboDrugGrowth_Nov2017); O'Neil et al. Merck screen.

Held-out pairs, 5 seeds.

EXTERNAL CHECK

The premise replicates on another lab's screen

1 Test 1

Rebuilt inside the Merck / O'Neil screen

0.81 rank correlation, other-cancer synergy vs lung synergy (583 pairs)

90% top 20 precision: 90% hits against a 25% base rate ($p < 0.0001$)

100% on 5 lung lines never in NCI-ALMANAC: 100% vs 26%

2 Test 2

Our own model, on shared pairs

+0.27 rank correlation with Merck lung results ($p = 0.002$)

Weaker, and we say so: the top-10 enrichment (40% vs 19%) is not significant at $p = 0.087$.

The premise **replicates**; this model's cross-screen agreement is **modest**.

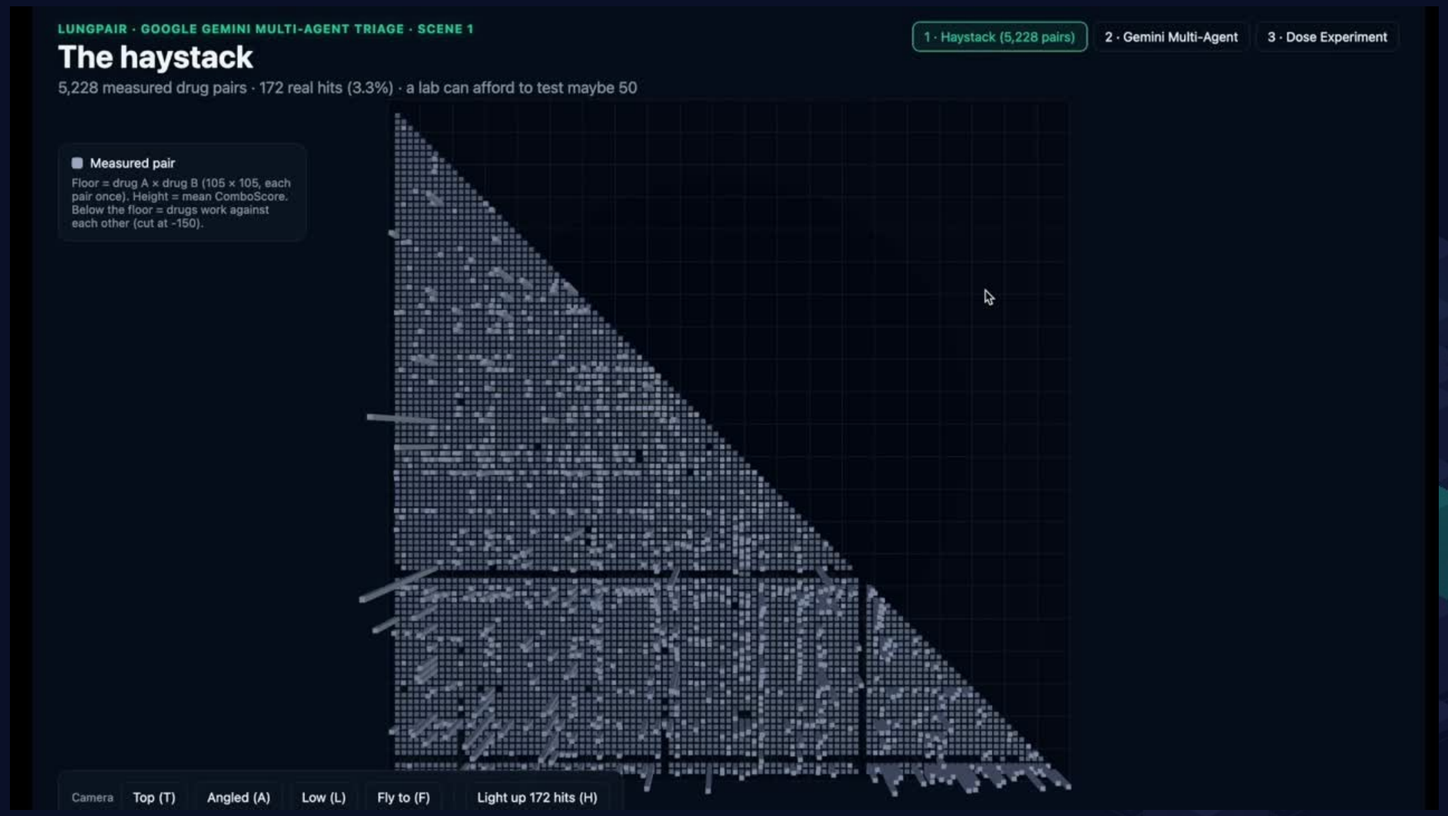
O'Neil et al. 2016 Merck screen via MTLSynergy.

Different lab, different synergy metric, different cell lines.

LIVE DEMO

Open this link: https://yonghwa11701-hash.github.io/lungpair-3d-visual/

The agent at work



1 3D landscape of every measured pair

2 A judge asks: which partner for gefitinib?

3 Agents rank, research and check

4 Researcher approves the batch; receipt saved

The 3D view replays measured NCI-ALMANAC results, not predictions.
The agent runs live on public cell-line data. No patient data.

THE BUSINESS CASE · ILLUSTRATIVE

Lower cost per real hit

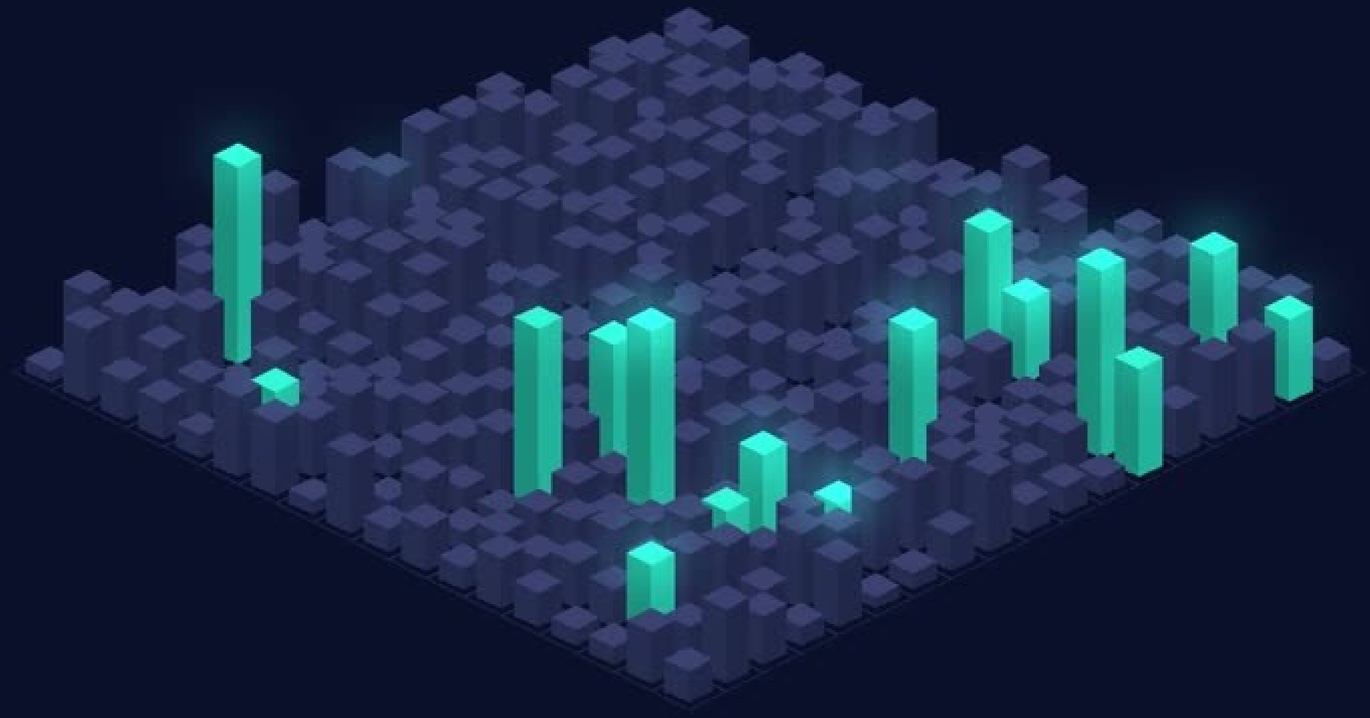
Strategy	Pairs	Cost	Real hits	Cost per hit
Test every pair	5,228	$16.0M	172	$93,000
Random 50	50	$153,000	~1.7	$93,000
LungPair top 50 *(pair seen in another cancer)*	**50**	**$153,000**	**31**	**$4,900**
LungPair top 20 *(brand-new pairs)*	**20**	**$61,200**	**~7**	**$8,700**

18X cheaper per real hit than random selection, at the same budget.

Illustrative: $340 per experiment (one pair, one cell line, academic core rate), 9 lung lines per pair.
Hit counts from an earlier run, to be updated.

THE BUSINESS CASE · ILLUSTRATIVE

Lower cost per real hit



THE BUSINESS CASE · ILLUSTRATIVE

Gemini vs other LLM

Benchmark		Gemini 4 Argon	GPT-5.2 Astra	Claude Fable 5.1	Claude Opus 5.5
Knowledge work	Vals Index	**68.9%**	63.1%	65.8%	67.0%
	AutomationBench	**51.3%**	41.4%	31.4%	42.5%
	Vals Finance Bench v2	**65.4%**	53.5%	58.9%	58.6%
Agentic coding	Harvey's Legal Agent Benchmark	**19.6%**	5.4%	6.7%	3.8%
	DeepSWE v1.1	**77.9%**	74.1%	67.4%	74.2%
	FrontierSWE v2	**55.0%**	65.5%	56.3%	62.3%
	Vibe Code Bench	**91.9%**	89.6%	90.3%	90.3%
	Terminal-bench 4.0	**57.4%**	58.2%	57.9%	66.4%
	PostTrainBench	**45.3%**	44.3%	40.2%	49.3%
Science and math	Terminal-Bench Science 0.1	**57.6%**	68.1%	52.6%	63.3%
	LABBench 2	**88.8%**	85.4%	68.6%	73.1%
	LlamaBench	**76.0%**	72.0%	65.6%	69.6%
Long context	GrepWalks	**99.7%**	98.7%	91.4%	90.6%
	LongBench v2	**84.2%**	71.8%	65.0%	66.8%
Computer use	OSWorld-2.0	**39.5%**	34.2%	—	38.2%
	Chartergraph	**69.2%**	72.6%	—	—
Multimodal understanding	LVGLBench	**91.7%**	87.5%	79.7%	83.7%
Cybersecurity	CWE-bench v1	**68.0%**	68.0%	58.0%	67.0%

Methodology: deepmind.google/models/evals-methodology/gemini-4-argon

REGULATORY AND SAFETY

Research tool now. Regulated tool later.

1

NOW

Research tool on public cell-line data.

No patient data.
Research use only.

↓

2

LATER · TO CONFIRM

Clinical decision support from blood ctDNA.

Likely regulated as software as a medical device; needs separate validation.

Risk	Mitigation
Read as treatment advice	Researcher-only scope; patient and dose questions refused before any model call.
Invented citations	Every PMID, DOI and URL must appear in the search grounding, or it is removed and flagged.
Invented or wrong numbers	Every number in the summary must exist in the ranker's output.
Confident guesses with no evidence	Tier 4 refused and listed as withheld; drugs outside the panel refused.
Agent acts without approval	Cannot order, buy or run anything; researcher signs off each batch; every call logged.

Gates are plain code, not a model. **A checker ‘pass’ never cancels a gate ‘fail’.**

IMPACT AND NEXT STEPS · ROADMAP, NOT BUILT

Re-rank before resistance sets in

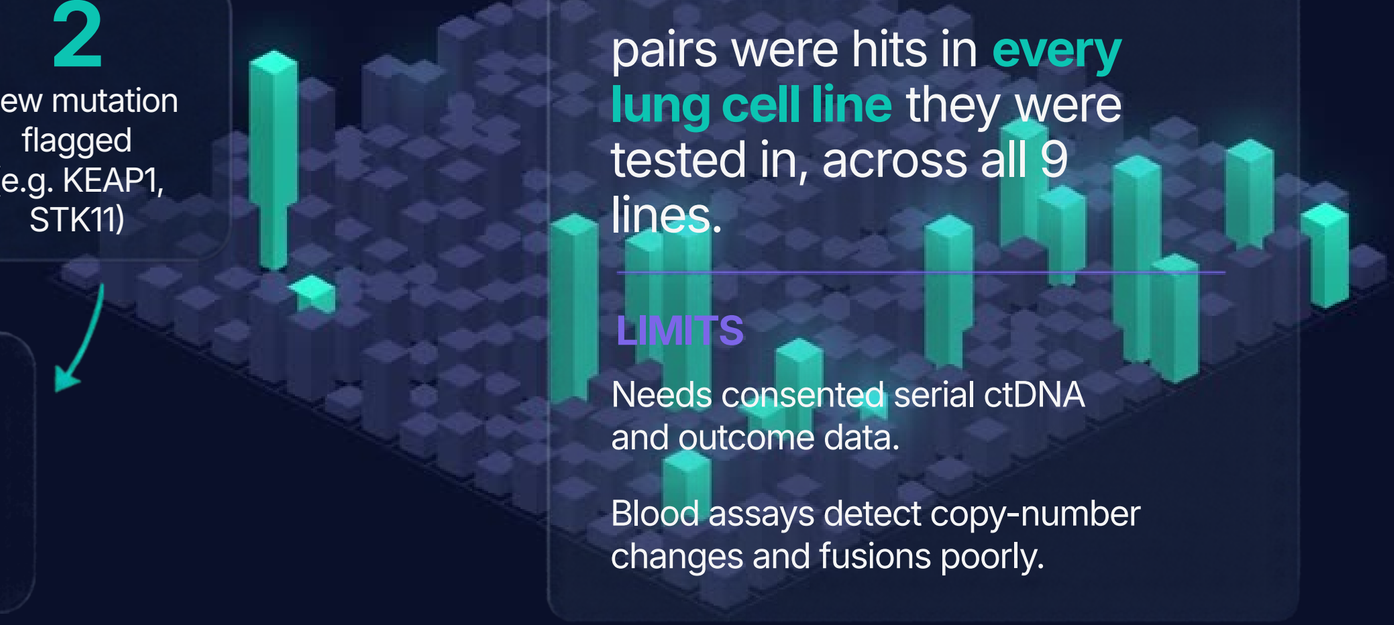
1 Blood ctDNA at baseline, then at set intervals

2 New mutation flagged (e.g. KEAP1, STK11)

3 LungPair re-ranks pairs by coverage

4 Clinician or researcher approves

5 Lab results feed the next ranking



WHY COVERAGE MATTERS

18

pairs were hits in **every lung cell line** they were tested in, across all 9 lines.

LIMITS

Needs consented serial ctDNA and outcome data.

Blood assays detect copy-number changes and fusions poorly.

LungPair. | **Fewer experiments, more real hits.**

Pascual et al., Ann Oncol 2022 (ESMO ctDNA recommendations); Kulkarni et al., 2023.

Market Scale, Profitability & B2B Expansion

$1.4B

TOTAL ADDRESSABLE MARKET

Global preclinical oncology screening & lead optimization software market

92%

SOFTWARE GROSS MARGIN

$0.08 query cost vs. $153,000 per physical screening batch

$15.5M

YEAR 4 TARGET ARR

64% Net EBITDA margin across 220 academic cores & 65 biopharma clients

1. Beachhead (Immediate • SOM: $12.5M)

- Customer: 250 Academic Cancer Screening Cores & Translation Institutes
- Pricing: $35,000 / year / facility (5 seats, unlimited triage)
- Budget: Departmental research consumable recharge account
- Advantage: ZERO hospital clinical procurement drag (4–8 week sales cycle)
- ROI: Saves $147,000 in reagents per 50-pair batch (>4x annual licence)

2. Near-Term Expansion (SAM: $132M)

- Customer: 1,100 Oncology Biopharma & Preclinical CROs (Charles River, WuXi)
- Pricing: $120,000 / year / pipeline licence + custom compound integration
- Value: Pre-filters combination leads before animal in vivo toxicity testing
- Unit Economics: <$0.025 marginal query cost yields 88%+ software margin
- Payback: Customer payback < 4.2 months (LTV / CAC = 33.5x)

3. Longer-Term Horizon (TAM: $1.4B)

- Customer: Hospital Oncology Centers & Commercial Liquid Biopsy Labs
- Pricing: $500 / patient test (Reimbursed via NHS & private healthcare)
- Technology: Serial blood ctDNA decision support (SaMD)
- Clinical Impact: Re-ranking combinations before clinical resistance emerges
- NICE Alignment: Ensures doublet therapies launch under £30k/QALY threshold

STRICTLY B2B SAAS MODEL: Zero hospital procurement drag • Sells out of consumable lab recharge budgets • 4-Year Scale: $540k ARR → $15.5M ARR